closed – form

explicit – form

enumerative combinatorics
graph theory

$P(n,k)$ = # of ways to select k out of n with order

$= n\cdot(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}$

$C(n,k)$ = # of ways to select k out of n without order

$\parallel$

$\binom{n}{k} = \frac{n!}{(n-k)!}\cdot\frac{1}{k!}$

Circular arrangement,

# of ways to arrange k out of n on a circle with order

$= \frac{P(n,k)}{k} = \frac{n!}{k\cdot(n-k)!}$

Multiple choice # $\binom{n}{t,\ n-t}$

$\binom{n}{k_1, k_2, \ldots, k_m} = \frac{n!}{k_1!\,k_2!\cdots k_m!} = \binom{n}{k_1}\binom{n-k_1}{k_2}\cdots\binom{k_m}{k_m}$

$\left(\sum_{i=1}^{m} k_i = n\right)$

Eg. Decide # of ways to arrange 10 couples on a table.

a. $2^{10} \cdot \frac{10!}{10}$

Eg. Decide # of ways to arrange 3 girls and 12 boys on a line

with girls non adjacent & not on ends

a. $C_{11}^{3} \cdot 3! \cdot 12!$

$[n] = \{1, 2, \dots, n\}$

Eg find : # of ways to partition $[n]$ into b_1 sets of size 1

$\vdots$

b_k sets of size k.

with $\sum_{i=1}^{k} i b_i = n$

Sol: $\binom{n}{\underbrace{1\cdots1}_{b_1}\underbrace{2\cdots2}_{b_2}\cdots\underbrace{k\cdots k}_{b_k}} \frac{1}{b_1!} \cdot \frac{1}{b_2!} \cdots \frac{1}{b_k!}$

Eg : $\sigma \in S_n$, $\text{type}(\sigma) = (l_1(\sigma), l_2(\sigma), \dots, l_n(\sigma))$

$l_i(\sigma) =$ # of i-cycles in the cyclic representation of σ

Given $(l_1, \dots l_n)$. find the # of $\delta \in S_n$ with $\text{type}(\delta) = (l_1, \dots, l_n)$

Sol: $\frac{n!}{\prod_{i=1}^{n} i^{l_i} \cdot (l_i)!}$

Given k types of objects. # of ways to select n.

<u>Sol</u>: $x_1 + x_2 + \dots + x_k = n \quad (x_i \in \mathbb{N})$

$y_1 + \dots + y_k = n + k \quad (y_i \in \mathbb{Z}^+)$

$$C_{n+k-1}^{k-1}$$

<u>Eg</u>: # of ways to take 12 out of 6 types, each type ≥ 1 ?

$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 12$, C_{11}^{5}

<u>The Twelve-fold Country</u>